

$A, B, C, \dots$, and the values $\xi, \xi_1, \xi_2, \dots$ by points $P, P_1, P_2, \dots$ the first of which is assumed at pleasure, and the others each from the preceding one by the like given geometrical construction. The problem is to determine the regions of the plane such that, P being taken at pleasure anywhere within one region, we arrive ultimately at the point A ; anywhere within another region at the point B ; and so for the several points representing the roots of the equation.

The solution is easy and elegant in the case of a quadric equation: but the next succeeding case of the cubic equation appears to present considerable difficulty.

Cambridge, March 3rd, 1879.

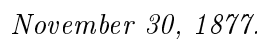
NO. 4. THE MECHANICAL CONSTRUCTION OF CONFORMABLE FIGURES.

[From the *American Journal of Mathematics*, t. ii. (1879), p. 186.]

Is it possible to devise an apparatus for the mechanical construction of conformable figures; that is, figures which are similar as regards corresponding infinitesimal areas? The problem is to connect mechanically two points P_1 and P_2 in such wise that P_1 (1) shall have two degrees of freedom (or be capable of moving over a plane area) its position always determining that of P_2 : (2) that if P_1, P_2 describe the infinitesimal lengths P_1Q_1, P_2Q_2 , then the ratio of these lengths, and their mutual inclination, shall depend upon the position of P_1 , but be independent of the direction of P_1Q_1 : or what is the same thing, that if P_1 describe uniformly an indefinitely small circle, then P_2 shall also describe uniformly an indefinitely small circle, the ratio of the radii, and the relative position of the starting points in the two circles respectively, depending on the position of P_1 .

Of course a pentagraph is a solution, but the two figures are in this case similar; and this is not what is wanted. Any unadjustable apparatus would give one solution only: the complete solution would be by an apparatus containing, suppose, a flexible lamina, so that P_1 moving in a given right line, the path of P_2 could be made to be any given curve whatever.

Cambridge, July 9th, 1879.



696.

CALCULATION OF THE MINIMUM N.G.F. OF THE BINARY SEVENTHIC.

[From the *American Journal of Mathematics*, t. ii. (1879), pp. 71–84.]

For the binary seventhic $(a, \dots | x, y)^7$ the number of the asyzygetic covariants $(\alpha, \dots)^\theta(x, y)^\mu$, or say of the deg-order $(\theta \cdot \mu)$, is given as the coefficient of $a^\theta x^\mu$ in the function

$$\frac{1 - x^{-1}}{1 - ax^7 \cdot 1 - ax^5 \cdot 1 - ax^3 \cdot 1 - ax \cdot 1 - ax^{-1} \cdot 1 - ax^{-3} \cdot 1 - ax^{-5} \cdot 1 - ax^{-7}}$$

developed in ascending powers of a . See my “Ninth Memoir on Quantics,” *Phil. Trans.*, t. CLXI. (1871), pp. 17–50, [462].

This function is in fact

$$= A(x) - \frac{1}{x^2} A\left(\frac{1}{x}\right),$$

where, developing in ascending powers of a , the second term $-\frac{1}{x^2} A\left(\frac{1}{x}\right)$ contains only negative powers of x , and it may consequently be disregarded: the number of asyzygetic covariants of the deg-order $(\theta \cdot \mu)$ is thus equal to the coefficient of $a^\theta x^\mu$ in the function $A(x)$, which function is for this reason called the Numerical Generating Function (N.G.F.) of the binary seventhic; and the function $A(x)$ expressed as a fraction in its least terms is said to be the minimum N.G.F.

According to a theorem of Professor Sylvester’s (*Proc. Royal Soc.*, t. XXVIII. (1878), pp. 11–13), this minimum N.G.F. is of the form

$$\frac{Z_0 + aZ_1 + a^2Z_2 + \dots + a^{36}Z_{36}}{1 - ax \cdot 1 - ax^3 \cdot 1 - ax^5 \cdot 1 - ax^7 \cdot 1 - a^4 \cdot 1 - a^8 \cdot 1 - a^{12} \cdot 1 - a^{20}},$$

where $Z_0, Z_1, \dots, Z_{36}$ are rational and integral functions of x of degrees not exceeding 14; and where, as will presently be seen, there is a symmetry in regard to the terms $Z_0, Z_{36}; Z_1, Z_{35};$ &c., equidistant from the middle term Z_{18} , such that the terms $Z_0, \dots, Z_{18}$ being known, the remaining terms $Z_{19}, \dots, Z_{36}$ can be at once written down.

Using only the foregoing properties, I obtained for the N.G.F. an expression which I communicated to Professor Sylvester, and which is published, *Comptes Rendus*, t. LXXXVII. (1878), p. 505, but with an erroneous value for the coefficient of a^7 and for that of the corresponding term a^{29} .¹ The correct value is Numerator of Minimum N.G.F. is =

$$\begin{aligned}
 &1 \\
 &+ a(-x - x^3 - x^5) \\
 &+ a^2(x^2 + x^4 + 2x^6 + x^8 + x^{10}) \\
 &+ a^3(-x^3 - x^5 - x^7 - x^{12}) \\
 &+ a^4(2x^4 + x^6 + x^{14}) \\
 &+ a^5(x + 3x^6 - x^8 - x^{10}) \\
 &+ a^6(-1 + 2x^4 - x^6 - x^8 + x^{10} - x^{14}) \\
 &+ a^7(4x + x^2 + 3x^6 - x^8 - x^{10}) \\
 &+ a^8(2 - x^2 - 3x^6 - 3x^8 - x^{10} - x^{14}) \\
 &+ a^9(x + 3x^2 + x^4 - x^6 + 2x^{10} + 2x^{12}) \\
 &+ a^{10}(-1 - 4x^2 - x^4 - 2x^6 - 3x^{10} - x^{14}) \\
 &+ a^{11}(x - 3x^2 + 2x^4 - x^6 - 2x^8 - x^{10} + x^{14}) \\
 &+ a^{12}(5 + x^2 - 4x^4 - 6x^6 - 4x^8 - x^{10} + 2x^{14}) \\
 &+ a^{13}(x - 4x^2 - 4x^4 - x^6 - x^8 - 4x^{12}) \\
 &+ a^{14}(2 - 3x^2 - x^4 - x^6 - 2x^8 - 3x^{10} - x^{14}) \\
 &+ a^{15}(3x - x^2 - x^4 - 7x^6 - 5x^8 - x^{10} - x^{12}) \\
 &+ a^{16}(6 + 3x^2 - 3x^4 - 4x^6 - 3x^8 - x^{10} - 5x^{14}) \\
 &+ a^{17}(-x - 2x^2 - 6x^4 - 8x^6 - 4x^8 - 3x^{10} - 4x^{14}) \\
 &+ a^{18}(2 + 6x^2 + x^4 + 2x^6 + 2x^8 + x^{10} + 6x^{12} + 2x^{14}) \\
 &+ a^{19}(4x - 3x^2 - 4x^4 - 8x^6 - 9x^8 - 2x^{10} - x^{14}) \\
 &+ a^{20}(5 - x^2 - 3x^4 - 4x^6 - 8x^8 - 3x^{10} - 6x^{14}) \\
 &+ a^{21}(-x - x^4 - 5x^6 - 7x^8 - x^{10} - x^{12} - 3x^{14}) \\
 &+ a^{22}(-1 - 3x^2 - 2x^4 - x^6 + x^{10} + 5x^{12} - x^{14}) \\
 &+ a^{23}(4x + x^2 - x^6 - 4x^8 - 4x^{10} + x^{14})
 \end{aligned}$$

¹The existence of these errors was pointed out to me by Professor Sylvester in a letter dated 13th November, 1878.

$$\begin{aligned}
 &+ a^{24} (2 - x^4 - 4x^6 - 6x^8 - 4x^{10} + x^{14}) \\
 &+ a^{25} (x - x^2 - 2x^6 - x^8 + 2x^{10} + 3x^{12} + 5x^{14}) \\
 &+ a^{26} (-1 - 2x^4 - 2x^6 - x^8 + 4x^{10} - x^{14}) \\
 &+ a^{27} (2x + 2x^2 - x^4 - x^6 + x^8 + 4x^{10} - x^{14}) \\
 &+ a^{28} (-x - x^4 - 3x^6 - 3x^8 - x^{10} - 2x^{14}) \\
 &+ a^{29} (x^2 - x^4 - 3x^6 - x^8 - 4x^{14}) \\
 &+ a^{30} (-x^6 + x^{10} + 2x^{12} + 3x^{14}) \\
 &+ a^{31} (1 + x^4 + 2x^{14}) \\
 &+ a^{32} (-x - x^2 - x^4 - x^6) \\
 &+ a^{33} (x^2 + x^4 + 2x^6 + x^{10} + x^{14}) \\
 &+ a^{34} (-x^6 - x^{12} - x^{14}) \\
 &+ a^{35} \cdot x^{14}.
 \end{aligned}$$

Denominator (as mentioned before) is

$$= 1 - ax \cdot 1 - ax^3 \cdot 1 - ax^5 \cdot 1 - ax^7 \cdot 1 - a^4 \cdot 1 - a^8 \cdot 1 - a^{12} \cdot 1 - a^{20}.$$

The method of calculation is as follows: write for a moment

$$\phi(a, x) = \frac{1 - x^{-1}}{1 - ax^7 \cdot 1 - ax^5 \cdot 1 - ax^3 \cdot 1 - ax \cdot 1 - ax^{-1} \cdot 1 - ax^{-3} \cdot 1 - ax^{-5} \cdot 1 - ax^{-7}},$$

then $\phi(a, x)$, developed in ascending powers of a , and rejecting from the result all negative powers of x , is

$$= \frac{Z_0 + aZ_1 + \cdots + a^{36}Z_{36}}{1 - ax \cdot 1 - ax^3 \cdot 1 - ax^5 \cdot 1 - ax^7 \cdot 1 - a^4 \cdot 1 - a^8 \cdot 1 - a^{12} \cdot 1 - a^{20}}$$

developed in like manner in ascending powers of a ; for the determination of the Z 's up to Z_{18} we require only the development of $\phi(a, x)$ up to a^{18} ; and, assuming that each Z is at most of the degree 14 in x , we require the coefficients of the different powers of a in $\phi(a, x)$ only up to x^{14} . Assuming then that $\phi(a, x)$ developed in ascending powers of a , up to a^{18} , rejecting all negative powers of x , and all positive powers greater than x^{14} , is

$$= X_0 + aX_1 + \cdots + a^{18}X_{18},$$

we have

$$X_0 + aX_1 + \cdots + a^{18}X_{18} = \frac{Z_0 + aZ_1 + \cdots + a^{18}Z_{18}}{1 - ax \cdot 1 - ax^3 \cdot 1 - ax^5 \cdot 1 - ax^7 \cdot 1 - a^4 \cdot 1 - a^8 \cdot 1 - a^{12} \cdot 1 - a^{20}},$$

or say

$$Z_0 + aZ_1 + \cdots + a^{18}Z_{18} = 1 - a^4 \cdot 1 - a^8 \cdot 1 - a^{12} \cdot 1 - a^{20} \cdot 1 - ax \cdot 1 - ax^3 \cdot 1 - ax^5 \cdot 1 - ax^7 (X_0 + aX_1 + \cdots + a^{18}X_{18});$$

viz. developing here the right-hand side as far as a^{18} , but in each term rejecting the powers of x above x^{14} , the coefficients of the several powers $a^0, a^1, a^2, \dots$, give the

required values $Z_0, Z_1, \dots, Z_{18}$. We require, therefore, only to know the values of these functions $X_0, X_1, \dots, X_{18}$.

To make a break in the calculation, it is convenient to write

$$1 - ax \cdot 1 - ax^3 \cdot 1 - ax^5 \cdot 1 - ax^7 (X_0 + aX_1 + \dots + a^{18}X_{18}) = Y_0 + aY_1 + \dots + a^{18}Y_{18};$$

putting then

$$1 - ax \cdot 1 - ax^3 \cdot 1 - ax^5 \cdot 1 - ax^7 = 1 - ap + a^2q - a^3r,$$

where (up to x^{14})

$$\begin{aligned} p &= x + x^3 + x^5 + x^7, \\ q &= x^4 + x^6 + 2x^8 + x^{10} + x^{12}, \\ r &= x^9 + x^{11} + x^{13}, \end{aligned}$$

we have

$$Y_0 + aY_1 + a^2Y_2 + \dots + a^{18}Y_{18} = (1 - ap + a^2q - a^3r)(X_0 + aX_1 + a^2X_2 + \dots + a^{18}X_{18}).$$

The values of $Y_0, Y_1, \dots, Y_{18}$ then are

	Y_0	Y_1	Y_2	Y_3	$\dots$	Y_{18}
	X_0	X_1	X_2	X_3		X_{18}
		$-pX_0$	$-pX_1$	$-pX_2$		$-pX_{17}$
			$+qX_0$	$+qX_1$		$+qX_{16}$
				$-rX_0$		$-rX_{15}$

the values being taken to x^{14} only; and we then have

$$Z_0 + aZ_1 + \dots + a^{18}Z_{18} = 1 - a^4 \cdot 1 - a^8 \cdot 1 - a^{12} \cdot 1 - a^{20} (Y_0 + aY_1 + \dots + a^{18}Y_{18});$$

viz. the values of $Z_0, Z_1, \dots, Z_{18}$ are obtained from those of the Y 's by adjoining the further $-Y_{k-4}, -Y_{k-8}, -Y_{k-12}, -Y_{k-20}$ terms (with their pairwise products $+Y_{k-12}, +2Y_{k-20}$ etc.) as set out at length in the original table.

The rule of symmetry, before referred to, is that the coefficient Z_{18+p} of a^{18+p} is obtained from the coefficient Z_p of a^p by changing each power x^q into x^{14-q} , the coefficients being unaltered; in particular Z_{18} , the coefficient of a^{18} , must remain

unaltered when each power x^q is changed into x^{14-q} ; and the verification thus obtained of the value

$$2Z_{18} = 2 + 6x^2 + x^4 + 2x^6 + 2x^8 + x^{10} + 6x^{12} + 2x^{14}$$

is in fact almost a complete verification of the whole work. Some other verifications, which present themselves in the course of the work, will be referred to further on.

We have, therefore, to calculate the coefficients $X_0, X_1, \dots, X_{18}$; the function $\phi(a, x)$ regarded as a function of a is at once decomposed into simple fractions; viz. we have

$$\begin{aligned} \phi(a, x) &= \frac{1 - x^{-1}}{1 - ax^7 \cdot 1 - ax^5 \cdot 1 - ax^3 \cdot 1 - ax \cdot 1 - ax^{-1} \cdot 1 - ax^{-3} \cdot 1 - ax^{-5} \cdot 1 - ax^{-7}} \\ &= \frac{x^{56}}{1 - x^2 \cdot 1 - x^4 \cdot 1 - x^6 \cdot 1 - x^8 \cdot 1 - x^{10} \cdot 1 - x^{12} \cdot 1 - x^{14}} \cdot \frac{1}{1 - ax^7} \\ &\quad - \frac{x^{40}}{1 - x^2 \cdot (1 - x^4)^2 \cdot 1 - x^6 \cdot 1 - x^8 \cdot 1 - x^{10}} \cdot \frac{1}{1 - ax^5} \\ &\quad + \frac{x^{24}}{1 - x^2 \cdot (1 - x^4)^2 \cdot (1 - x^6)^2 \cdot 1 - x^8} \cdot \frac{1}{1 - ax^3} \\ &\quad - \frac{x^8}{(1 - x^2) \cdot (1 - x^4)^2 \cdot (1 - x^6)^2 \cdot 1 - x^8} \cdot \frac{1}{1 - ax} \end{aligned}$$

(the terms in $\frac{1}{1 - ax^{-1}}, \dots, \frac{1}{1 - ax^{-7}}$ produce only negative powers of x and are accordingly discarded).

In order to obtain the expansion of $\phi(a, x)$ in the assumed form of an expansion in ascending powers of a , we must of course expand the simple fractions $\frac{1}{1 - ax^k}$, &c., in ascending powers of a , but it requires a little consideration to see that we must also expand the x -coefficients of these simple fractions in ascending powers of x . For instance, as regards the term independent of a , here developing the several coefficients as far as x^{14} , the last five terms give (see post)

$$\begin{aligned} &+ x^8 + x^{10} + 3x^{12} + 5x^{14} + 9x^{16} \\ &- x^6 - x^8 - 3x^{10} - 5x^{12} - 8x^{14} - 11x^{16} - 15x^{18} - 24x^{20} \\ &+ 1 + x^2 + 2x^4 + 3x^6 + 5x^8 + 7x^{10} + 11x^{12} + 14x^{14} + 20x^{16} + 26x^{18} \\ &- x^4 - x^6 - x^8 - 2x^{10} - 2x^{12} - 4x^{14} - 4x^{16} - 6x^{18} - 7x^{20} - 10x^{22} \\ &= -x^{-2} + 1 \end{aligned}$$

viz. the sum is $= -1 \cdot x^{-2}$ as it should be.²

²To give the last degree of perfection to the beautiful method of Professor Cayley it would seem desirable that a proof should be given of the principle illustrated by the example in the text, and the nature of the mischief resulting from its neglect clearly pointed out.—EDS. OF THE A. J. M.

The expansion is required only as far as x^{14} : the first four terms are therefore to be disregarded, and, writing for shortness

$$\begin{aligned} E &= \frac{1}{1 - x^2 \cdot (1 - x^4)^2 \cdot (1 - x^6)^2 \cdot 1 - x^8}, \\ F &= \frac{1}{1 - x^2 \cdot (1 - x^4)^2 \cdot 1 - x^6 \cdot 1 - x^8 \cdot 1 - x^{10}}, \\ G &= \frac{1}{1 - x^2 \cdot 1 - x^4 \cdot 1 - x^6 \cdot 1 - x^8 \cdot 1 - x^{10} \cdot 1 - x^{12}}, \\ H &= \frac{1}{1 - x^2 \cdot 1 - x^4 \cdot 1 - x^6 \cdot 1 - x^8 \cdot 1 - x^{10} \cdot 1 - x^{12} \cdot 1 - x^{14}}, \end{aligned}$$

we have

$$\phi(a, x) = \frac{x^8 E}{1 - ax} - \frac{x^{24} F}{1 - ax^3} + \frac{x^{40} G}{1 - ax^5} - \frac{x^{56} H}{1 - ax^7},$$

which is

$$\begin{aligned} &= x^8 E (1 + ax^{-1} + a^2 x^{-2} + \dots) \\ &\quad - x^{24} F (1 + ax^{-3} + a^2 x^{-6} + \dots) \\ &\quad + x^{40} G (1 + ax^{-5} + a^2 x^{-10} + \dots) \\ &\quad - x^{56} H (1 + ax^{-7} + a^2 x^{-14} + \dots), \end{aligned}$$

where the several series are to be continued up to a^{18} , and, after substituting for E, F, G, H their expansions in ascending powers of x , we are to reject negative powers of x , and also powers higher than x^{14} . The functions E, F, G, H contain each of them only even powers of x , and it is easy to see that we require the expansions up to x^{32}, x^{64}, x^{96} and x^{128} respectively. For the sake of a verification, I in fact calculated E, F up to x^{64} and G, H up to x^{142} : viz. we have

$$(1 - x^{10})E = (1 - x^8)F,$$

from the coefficients of E we have those of $(1 - x^{10})E$, and in the process of calculating F we have at the last step but one the coefficients of $(1 - x^8)F$, the agreement of the two sets being the verification; similarly,

$$(1 - x^{12})G = (1 - x^6)H$$

gives a verification. The process for the calculation of E ,

$$\frac{1}{1 - x^2 \cdot (1 - x^4)^2 \cdot (1 - x^6)^2 \cdot 1 - x^8},$$

is as follows:

Ind. x	0	2	4	6	8	10	12	14	16	18	20	22
$(1 - x^2)^{-1}$	1	1	1	1	1	1	1	1	1	1	1	1
		1	1	2	2	3	3	4	4	5	5	5
$(1 - x^4)^{-1}$	1	1	2	2	3	3	4	4	5	5	6	6
		1	1	3	3	6	6	10	10	15	15	21
$(1 - x^4)^{-1}$	1	1	3	3	6	6	10	10	15	15	21	21
		1	1	3	4	7	9	14	17	24	29	38
$(1 - x^6)^{-1}$	1	1	3	4	7	9	14	17	24	29	38	45
		1	1	3	5	8	12	19	25	36	48	63
$(1 - x^6)^{-1}$	1	1	3	5	8	12	19	25	36	48	63	81
		1	1	3	5	9	13	22	30	45	61	85
$E = (1 - x^8)^{-1}$	1	1	3	5	9	13	22	30	45	61	85	111

the alternate lines giving the developments of the functions

$$(1 - x^2)^{-1}, (1 - x^4)^{-2}, (1 - x^4)^{-2}(1 - x^6)^{-1}, (1 - x^4)^{-2}(1 - x^6)^{-2}, \dots,$$

which are the products of the x -functions down to any particular line. And in like manner we have the expansions of the other functions F, G, H respectively. I give first the expansions of E, F, G, H ; next the calculation of the X 's; then the calculation of the Y 's: and from these the Z 's up to Z_{18} , or coefficients of the powers $a^0, a^1, \dots, a^{18}$ in the numerator of the N.G.F. are at once found; and the coefficients of the remaining powers $a^{19} \dots a^{36}$ are then deduced, as already mentioned.

Writing in the formula $x = 0$, we have, for the numerator of the N.G.F. of the invariants, the expression

$$1 - a^4 + 2a^8 - a^{12} + 5a^{16} + 2a^{18} + 6a^{20} + 2a^{22} + 5a^{24} - a^{28} + 2a^{32} - a^{36} + a^{40}$$

agreeing with a result in my "Second Memoir on Quantics," *Phil. Trans.*, t. CXLVI. (1856), [Number 141, vol. II. in this Collection, p. 266]; this, then, was a known result, and it affords a verification, not only of the terms in a^0 , but also of those in x^0 . Thus, in calculating the foregoing expression of the numerator, we obtain $Z_4 = +(2x^4 + x^6 + x^{14})$, viz. the term is

$$a^4(2x^4 + x^6 + x^{14}),$$

and we thence have the corresponding term $a^{34}(1 + x^8 + 2x^{10})$, which, when $x = 0$, becomes $= a^{34}$, a term of the numerator for the invariants: and the term $1x^{14}$ of Z_4

is thus verified, viz. so soon as, in the calculation, we arrive at this term, we know that it is right, and the calculation up to this point is, to a considerable extent, verified. And similarly, in continuing the calculation, we arrive at other terms which are verified in the like manner.

Expansions of the Functions E, F, G, H.

Ind. <i>x</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	Ind. <i>x</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	Ind. <i>x</i>	<i>H</i>
0	1	1	1	1	16	45	36	20	6	108	2265
2	1	1	1		18	61	47	26	7	110	2426
4	3	3	2	1	20	85	66	35	10	112	2623
6	5	4	3	1	22	111	84	44	11	114	2807
8	9	8	5	2	24		113	58	16	116	3026
10	13	11	7	2	26		141	71	17	118	3232
12	22	18	11	4	28		183	90	23	120	3479
14	30	24	14	4	30		225	110	26	122	3708

Ind. <i>x</i>	<i>F</i>	<i>G</i>	<i>H</i>	Ind. <i>x</i>	<i>G</i>	<i>H</i>	Ind. <i>x</i>	<i>H</i>
32	284	136	33	70	2172	419	124	3981
34	344	163	37	72	2432	472	126	4240
36	425	199	47	74	2702	515	128	4541
38	508	235	52	76	3009	576	130	4828
40	617	282	64	78	3331	629	132	5164
42	729	331	72	80	3692	699	134	5481
44	872	391	86	82	4070	760	136	5850
46	1020	454	96	84	4494	843	138	6204
48	1205	532	115	86	4935	913	140	6609
50	1397	612	127	88	5427	1007	142	6998
52	1632	709	149	90	5942	1091		
54	1877	811	166	92	6510	1197		
56	2172	931	192	94	7104	1293		
58	2480	1057	212	96	7760	1416		
60	2846	1206	245	98	8442	1525		
62	3228	1360	269	100	9192	1663		
64	3677	1540	307	102	9975	1790		
66		1729	338	104	10829	1945		
68		1945	382	106		2088		

Calculation of the X 's.

Ind. x even or odd according as suffix X is even or odd.

(Detailed columnar layout in the source: each block lists the partial-fraction contributions in the columns $0_1, 2_3, 4_5, 6_7, 8_9, 10_{11}, 12_{13}, 14$, and the row sums give $X_0, X_1, \dots, X_{18}$.)

	0 ₁	2 ₃	4 ₅	6 ₇	8 ₉	10 ₁₁	12 ₁₃	14
			−1	−1	−3	1	1	3
					−4	−8		−11
	1	1	2	3	5	7	11	14
		−1	−1	−2	−2	−4	−4	−6
$X_0 =$	1							
				1	1	3		
	−1	−1	−3	−4	−8	−11	−18	
	3	5	7	11	14	20	26	
	−2	−4	−4	−6	−7	−10	−11	
$X_1 =$				+1				
	−1	−3	−4	−8		1	1	3
	5	7	11	14	20	−11	−18	−24
	−6	−7	−10	−11	26	35	44	58
		+1		+1	−16	−17	−23	−26
$X_2 =$						+1		+1
				1	1	3	5	9
	−4	−8	−11	−18	−24	−36	−47	−66
	20	26	35	44	58	71	90	110
	−16	−17	−23	−26	−33	−37	−47	−52
$X_3 =$			+1	+1	+1	+2	+1	+1

(Further blocks for X_4, X_5, X_6 on this page; the values obtained are):

$$X_4 = 1; \quad X_5 = 1 + 0 + 3(\cdot); \quad X_6 = 1 + 2(\cdot).$$

[The full ledger continues with several lines per X_k giving the contributions $-x^8E, +x^{24}F, -x^{40}G, +x^{56}H$ for each power of x .]

	0 ₁	2 ₃	4 ₅	6 ₇	8 ₉	10 ₁₁	12 ₁₃	14
		1		1	3	5	9	13
	−24	−36	−47	−66	−84	−113	−141	−183
	110	136	163	199	235	282	331	391
	−86	−96	−115	−127	−149	−166	−191	−212
X ₇ =	3+	4+	4+	7+	9+	10+	11+	13
		1	1	3	5	9	13	22
	−66	−84	−113	−141	−183	−225	−284	−344
	282	331	391	454	532	612	709	811
	−212	−245	−269	−307	−338	−382	−419	−472
X ₈ =	4+	3+	10+	9+	16+	14+	19+	17
	1	1	3	5	9	13	22	
	−113	−141	−183	−225	−284	−344	−425	
	454	532	612	709	811	931	1057	
	−338	−382	−419	−472	−515	−576	−629	
X ₉ =	4+	10+	13+	17+	21+	24+	25	
		1	1	3	5	9	13	22
	−141	−183	−225	−284	−344	−425	−508	−617
	612	709	811	931	1057	1206	1360	1540
	−472	−515	−576	−629	−699	−760	−843	−913
X ₁₀ =		+12	+13	+23	+23	+34	+31	+40
	1	3	5	9	13	22	30	
	−225	−284	−344	−425	−508	−617	−729	
	931	1057	1206	1360	1540	1729	1945	
	−699	−760	−843	−913	−1007	−1091	−1197	
X ₁₁ =	8+	16+	24+	31+	38+	43+	49	
		1	3	5	9	13	22	30
	−284	−344	−425	−508	−617	−729	−872	−1020
	1206	1360	1540	1729	1945	2172	2432	2702
	−913	−1007	−1091	−1197	−1293	−1416	−1525	−1663
X ₁₂ =	10+	12+	29+	33+	48+	49+	65+	64

	0 ₁	2 ₃	4 ₅	6 ₇	8 ₉	10 ₁₁	12 ₁₃	14
$X_{13} =$	3	5	9	13	22	30	45	
	−425	−508	−617	−729	−872	−1020	−1205	
	1729	1945	2172	2432	2702	3009	3331	
	−1293	−1416	−1525	−1663	−1790	−1945	−2088	
	14+	26+	39+	53+	62+	74+	83	
$X_{14} =$	3	5	9	13	22	30	45	61
	−508	−617	−729	−872	−1020	−1205	−1397	−1632
	2172	2432	2702	3009	3331	3692	4070	4494
	−1663	−1790	−1945	−2088	−2265	−2426	−2623	−2807
	4+	30+	37+	62+	68+	91+	95+	116
$X_{15} =$	5	9	13	22	30	45	61	
	−729	−872	−1020	−1205	−1397	−1632	−1877	
	3009	3331	3692	4070	4494	4935	5427	
	−2265	−2426	−2623	−2807	−3026	−3232	−3479	
	20+	42+	62+	80+	101+	116+	132	
$X_{16} =$	5	9	13	22	30	45	61	85
	−872	−1020	−1205	−1397	−1632	−1877	−2172	−2480
	3692	4070	4494	4935	5427	5942	6510	7104
	−2807	−3026	−3232	−3479	−3708	−3981	−4240	−4541
	18+	33+	70+	81+	117+	129+	159+	168
$X_{17} =$	9	13	22	30	45	61	85	
	−1205	−1397	−1632	−1877	−2172	−2480	−2846	
	4935	5427	5942	6510	7104	7760	8442	
	−3708	−3981	−4240	−4541	−4828	−5164	−5481	
	31+	62+	92+	122+	149+	177+	200	
$X_{18} =$	9	13	22	30	45	61	85	111
	−1397	−1632	−1877	−2172	−2480	−2846	−3228	−3677
	5942	6510	7104	7760	8442	9192	9975	10829
	−4541	−4828	−5164	−5481	−5850	−6204	−6609	−6998
	13+	63+	85+	137+	157+	203+	223+	265

Calculation of the Y 's.

Ind. x even or odd as suffix X is even or odd.

	0 ₁	2 ₃	4 ₅	6 ₇	8 ₉	10 ₁₁	12 ₁₃
$Y_0 =$	1						
$Y_1 =$	-1	-1	-1	-1			
$Y_2 =$	-1	-1	-1	0	0	0	0
	0	1	0	1	0	1	0
		-1	-1	-1	-1	-1	-1
		1	1	2	1	1	1
$Y_3 =$	0	1	1	1	1	1	0
		-1	-1	-2	-2	-2	-2
			-1		-1	-1	-1
$Y_4 =$	0	0	0	-1	-1	-1	-1
	1	0	3	1	3	2	3
		-1	-2	-3	-5	-5	-5
			1	1	3	2	4
$Y_5 =$	1	0	+2	0	+1		0
	1	2	3	4	4	5	4
	-1	-1	-4	-5	-7	-9	-9
		+1	-	1	2	4	6
$Y_6 =$	0	+1	-1	0	-1	-1	0
	4	2	4	7	5	8	7
	-1	-3	-7	-10	-13	-16	-17
		+1	-	5	5	11	10
$Y_7 =$	0	+3	0	+2	0	+1	0

	0_1	2_3	4_5	6_7	8_9	10_{11}	12_{13}	14
$Y_8 =$	3	4	7	9	10	11	13	
		-4	-6	-13	-18	-22	-27	
			1	3	7	12	17	
					-1	-1	-4	
	3	0	+2	-1	-2	0	-1	
$Y_9 =$	4	3	10	9	16	14	19	17
	-3	-7	-14	-23	-30	-37	-43	
		4	6	17	20	33		
			-1	-1	-3	-6		
	1	0	+1	+2	+3	+4		
$Y_{10} =$	0	+3	-1	-2	0	-2	+1	
	12	13	23	23	34	31	49	
		-14	-27	-44	-61	-75	-87	
			4	7	21	29	52	61
						-3	-7	-14
$Y_{11} =$	0	+8	+3	+3	0	-	1	+1
$Y_{12} =$	8	16	24	31	38	43	49	0
	-12	-25	-48	-71	-93	-111		
		4	14	31	54	78		
				-4	-7	-17		
$Y_{13} =$	8	+4	+3	-3	-6	-3	-1	
	10	29	33	48	49	65	64	
		-8	-24	-65	-79	-100	-136	-161
			12	25	60	84	128	
					-4	-14	-27	-87
	10	+4	+5	-3	-6	-4	-1	+4

	0 ₁	2 ₃	4 ₅	6 ₇	8 ₉	10 ₁₁	12 ₁₃	14
	14	26	39	53	62	74	83	
	−10	−22	−51	−84	−122	−159	−195	
		8	24	56	95	141		
		−4	−7	−4	−12	−25		
					−12	−25		
Y ₁₄ =	4	+4	+8	+24	+56	+95	+141	
	20	30	37	62	68	91	95	116
	−14	−40	−79	−132	−180	−228	−272	
		16	22	61	96	161	204	
		+7	−3	−8	−24	−48		
Y ₁₅ =	4	+18	+16	+22	+18	+13	+13	
	−18	42	62	80	101	116	132	
	13	−34	−71	−133	−197	−258	−316	
		8	14	40	93	158	233	
		+5	−13	−10	−22	−51		
				−13	−6	−2		
Y ₁₆ =			+5					
	33	70	81	117	129	159	168	
	−20	−62	−124	−204	−285	−359	−429	
	+13	4	34	75	163	238	350	
		+12	−9	−12	−14	−40	−79	
					−7	−2	+10	
Y ₁₇ =	26	24	−18	−24	−14	−4	+20	
	62	92	122	149	177	200		
	−51	−121	−202	−301	−397	−486		
	+11	20	62	144	246	367		
		−9	−18	−4	−12	−34	−71	
				−12	−8	+10		
Y ₁₈ =	22	−18	−36	−24	+6	+57		
	63	85	137	157	203	223	265	
	−31	−93	−185	−307	−425	−540	−648	
	+32	18	51	139	235	389	511	
		+10	+3	−11	−20	−62	−124	
					−7	+10	+4	

Cambridge, December 7th, 1878.

697.

ON THE DOUBLE ϑ -FUNCTIONS.

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I HAVE sought to obtain, in forms which may be useful in regard to the theory of the double ϑ -functions, the integral of the elliptic differential equation

$$\frac{dx}{\sqrt{a-x \cdot b-x \cdot c-x \cdot d-x}} + \frac{dy}{\sqrt{a-y \cdot b-y \cdot c-y \cdot d-y}} = 0 :$$

the present paper has immediate reference only to this differential equation; but, on account of the design of the investigation, I have entitled it as above.

We may for the general integral of the above equation take a particular integral of the equation

$$\frac{dx}{\sqrt{a-x \cdot b-x \cdot c-x \cdot d-x}} + \frac{dy}{\sqrt{a-y \cdot b-y \cdot c-y \cdot d-y}} \pm \frac{dz}{\sqrt{a-z \cdot b-z \cdot c-z \cdot d-z}} = 0;$$

viz. this particular integral, regarding therein z as an arbitrary constant, will be the general integral of the first mentioned equation. And we may further assume that z is the value of y corresponding to the value a of x .

I write for shortness

$$\begin{aligned} a-x, b-x, c-x, d-x &= \mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{d}, \\ a-y, b-y, c-y, d-y &= \mathfrak{a}_1, \mathfrak{b}_1, \mathfrak{c}_1, \mathfrak{d}_1; \end{aligned}$$

and I write also (xy, bc, ad) , or more shortly (bc, ad) to denote the determinant

$$\begin{vmatrix} 1, & x+y, & xy \\ 1, & b+c, & bc \\ 1, & a+d, & ad \end{vmatrix}$$

we have of course $(ad, bc) = -(bc, ad)$, and there are thus the three distinct determinants (ad, bc) , (bd, ac) and (cd, ab) .

We have then for each of the functions

$$\sqrt{\frac{a-z}{d-z}}, \quad \sqrt{\frac{b-z}{d-z}}, \quad \sqrt{\frac{c-z}{d-z}}$$

a set of four equivalent expressions, the whole system being

$$\begin{aligned} \sqrt{\frac{a-z}{d-z}} &= \frac{\sqrt{a-b \cdot a-c} \{ \sqrt{a\bar{d}b_1c_1} + \sqrt{a_1\bar{d}_1bc} \}}{(bc, ad)} = \frac{\sqrt{a-b \cdot a-c} (x-y)}{\sqrt{a\bar{d}b_1c_1} - \sqrt{a_1\bar{d}_1bc}} \\ &= \frac{\sqrt{a-b \cdot a-c} \{ \sqrt{a\bar{b}c_1\bar{d}_1} + \sqrt{a_1\bar{b}_1c\bar{d}} \}}{(a-c)\sqrt{b\bar{d}b_1\bar{d}_1} - (b-d)\sqrt{a\bar{c}a_1c_1}} = \frac{\sqrt{a-b \cdot a-c} \{ \sqrt{a\bar{c}b_1\bar{d}_1} + \sqrt{a_1\bar{c}_1b\bar{d}} \}}{(a-b)\sqrt{c\bar{d}c_1\bar{d}_1} - (c-d)\sqrt{a\bar{b}a_1\bar{b}_1}} \\ \sqrt{\frac{b-z}{d-z}} &= \sqrt{-\frac{a-b}{a-d}} \frac{\{(a-c)\sqrt{b\bar{d}b_1\bar{d}_1} + (b-d)\sqrt{a\bar{c}a_1c_1}\}}{(bc, ad)} = \sqrt{-\frac{a-b}{a-d}} \frac{\{\sqrt{a\bar{b}c_1\bar{d}_1} - \sqrt{a_1\bar{b}_1c\bar{d}}\}}{\sqrt{a\bar{d}b_1c_1} - \sqrt{a_1\bar{d}_1bc}} \\ &= \sqrt{-\frac{a-b}{a-d}} \frac{(cd, ab)}{(a-c)\sqrt{b\bar{d}b_1\bar{d}_1} - (b-d)\sqrt{a\bar{c}a_1c_1}} = -\sqrt{-\frac{a-b}{a-d}} \frac{\{(a-d)\sqrt{b\bar{c}b_1c_1} + (b-c)\sqrt{a\bar{d}a_1\bar{d}_1}\}}{(a-b)\sqrt{c\bar{d}c_1\bar{d}_1} - (c-d)\sqrt{a\bar{b}a_1\bar{b}_1}} \\ \sqrt{\frac{c-z}{d-z}} &= -\sqrt{-\frac{a-c}{a-d}} \frac{\{(a-b)\sqrt{c\bar{d}c_1\bar{d}_1} + (c-d)\sqrt{a\bar{b}a_1\bar{b}_1}\}}{(bc, ad)} = -\sqrt{-\frac{a-c}{a-d}} \frac{\{\sqrt{a\bar{c}b_1\bar{d}_1} - \sqrt{a_1\bar{c}_1b\bar{d}}\}}{\sqrt{a\bar{d}b_1c_1} - \sqrt{a_1\bar{d}_1bc}} \\ &= -\sqrt{-\frac{a-c}{a-d}} \frac{\{(a-d)\sqrt{b\bar{c}b_1c_1} - (b-c)\sqrt{a\bar{d}a_1\bar{d}_1}\}}{(a-c)\sqrt{b\bar{d}b_1\bar{d}_1} - (b-d)\sqrt{a\bar{c}a_1c_1}} = \sqrt{-\frac{a-c}{a-d}} \frac{(bd, ac)}{(a-b)\sqrt{c\bar{d}c_1\bar{d}_1} - (c-d)\sqrt{a\bar{b}a_1\bar{b}_1}} \end{aligned}$$

The expressions in the like fourfold form for the functions $\text{sn}(u+v)$, $\text{cn}(u+v)$, $\text{dn}(u+v)$ are given p. 63 of my *Treatise on Elliptic Functions*.

It is easy to verify first that the four expressions for the same function of z are identical, and next that the expressions for the three several functions

$$\sqrt{\frac{a-z}{d-z}}, \quad \sqrt{\frac{b-z}{d-z}}, \quad \sqrt{\frac{c-z}{d-z}}$$

are consistent with each other. For instance, comparing the first and second expressions of $\sqrt{\frac{a-z}{d-z}}$, the equation to be verified is

$$= a\bar{d}b_1c_1 - a_1\bar{d}_1bc - (x-y)(bc, ad),$$

which is at once shown to be true. Again comparing the first and second expressions for $\sqrt{\frac{b-z}{d-z}}$, we ought to have

$$\{(a-c)\sqrt{b\bar{d}b_1\bar{d}_1} + (b-d)\sqrt{a\bar{c}a_1c_1}\} \{ \sqrt{a\bar{d}b_1c_1} - \sqrt{a_1\bar{d}_1bc} \} = (bc, ad) \{ \sqrt{a\bar{b}c_1\bar{d}_1} - \sqrt{a_1\bar{b}_1c\bar{d}} \}.$$

Here the product on the left-hand side is

$$= (a - c) \{ \mathfrak{b}_1 \mathfrak{d} \sqrt{\mathfrak{a} \mathfrak{b} \mathfrak{c}_1 \mathfrak{d}_1} - \mathfrak{b} \mathfrak{d}_1 \sqrt{\mathfrak{a}_1 \mathfrak{b}_1 \mathfrak{c} \mathfrak{d}} \} + (b - d) \{ -\mathfrak{a}_1 \mathfrak{c} \sqrt{\mathfrak{a} \mathfrak{b} \mathfrak{c}_1 \mathfrak{d}_1} + \mathfrak{a} \mathfrak{c}_1 \sqrt{\mathfrak{a}_1 \mathfrak{b}_1 \mathfrak{c} \mathfrak{d}} \},$$

viz. this is

$$= \sqrt{\mathfrak{a} \mathfrak{b} \mathfrak{c}_1 \mathfrak{d}_1} \{ (a - c) \mathfrak{b}_1 \mathfrak{d} - (b - d) \mathfrak{a}_1 \mathfrak{c} \} - \sqrt{\mathfrak{a}_1 \mathfrak{b}_1 \mathfrak{c} \mathfrak{d}} \{ (a - c) \mathfrak{b} \mathfrak{d}_1 - (b - d) \mathfrak{a} \mathfrak{c}_1 \},$$

and in this last expression the two terms in $\{ \}$ are at once shown to be each $= (bc, ad)$; whence the identity in question.

Comparing in like manner the first expressions for $\sqrt{\frac{b-z}{d-z}}$ and $\sqrt{\frac{c-z}{d-z}}$ respectively, we have

$$\begin{aligned} & - (b - d)(bc, ad)^2 \sqrt{-\frac{a-c}{a-d}} \frac{1}{z} \\ & = -(a - b)(a - c)(b - d) \{ \mathfrak{a} \mathfrak{d} \mathfrak{b}_1 \mathfrak{c}_1 + \mathfrak{a}_1 \mathfrak{d}_1 \mathfrak{b} \mathfrak{c} - 2 \sqrt{\mathfrak{a} \mathfrak{b} \mathfrak{c} \mathfrak{d} \mathfrak{a}_1 \mathfrak{b}_1 \mathfrak{c}_1 \mathfrak{d}_1} \}, \\ & - (d - a)(bc, ad)^2 \sqrt{-\frac{a-b}{a-d}} \frac{1}{z} \\ & = -(a - b) \{ (a - c)^2 \mathfrak{b} \mathfrak{d} \mathfrak{b}_1 \mathfrak{d}_1 + (b - d)^2 \mathfrak{a} \mathfrak{c} \mathfrak{a}_1 \mathfrak{c}_1 + 2(a - c)(b - d) \sqrt{\mathfrak{a} \mathfrak{b} \mathfrak{c} \mathfrak{d} \mathfrak{a}_1 \mathfrak{b}_1 \mathfrak{c}_1 \mathfrak{d}_1} \}, \end{aligned}$$

whence, adding, the radical on the right-hand side disappears; the whole equation divides by $-(a - b)$, and omitting this factor, the relation to be verified is

$$-(bc, ad)^2 = (a - c)^2 \mathfrak{b} \mathfrak{d} \mathfrak{b}_1 \mathfrak{d}_1 + (b - d)^2 \mathfrak{a} \mathfrak{c} \mathfrak{a}_1 \mathfrak{c}_1 - (a - c)(b - d) (\mathfrak{a} \mathfrak{d} \mathfrak{b}_1 \mathfrak{c}_1 + \mathfrak{a}_1 \mathfrak{d}_1 \mathfrak{b} \mathfrak{c})$$

the right-hand side is here

$$= -\{ (a - c) \mathfrak{b}_1 \mathfrak{d} - (b - d) \mathfrak{a}_1 \mathfrak{c} \} \{ (a - c) \mathfrak{b} \mathfrak{d}_1 - (b - d) \mathfrak{a} \mathfrak{c}_1 \},$$

and each of the two factors being $= (bc, ad)$, the identity is verified. It thus appears that the twelve equations are in fact equivalent to a single equation in x, y, z .

Writing in the several formulae $x = a, b, c, d$ successively, they become

$x = a,$	$x = b,$	$x = c,$	$x = d,$
$\frac{a - z}{d - z} = \frac{\mathfrak{a}_1}{\mathfrak{d}_1},$	$\frac{c - a \mathfrak{b}_1}{d - b \cdot \mathfrak{c}_1},$	$\frac{b - a \mathfrak{c}_1}{d - c \cdot \mathfrak{b}_1},$	$\frac{a - b \cdot a - c \cdot \mathfrak{d}_1}{d - b \cdot d - c \cdot \mathfrak{a}_1},$
$\frac{b - z}{d - z} = \frac{\mathfrak{b}_1}{\mathfrak{d}_1},$	$\frac{c - b \mathfrak{a}_1}{d - a \cdot \mathfrak{c}_1},$	$\frac{b - a \cdot b - c \mathfrak{d}_1}{d - a \cdot d - c \cdot \mathfrak{b}_1},$	$\frac{a - b \mathfrak{c}_1}{d - c \cdot \mathfrak{a}_1},$
$\frac{c - z}{d - z} = \frac{\mathfrak{c}_1}{\mathfrak{d}_1},$	$\frac{c - a \cdot c - b \mathfrak{d}_1}{d - a \cdot d - b \cdot \mathfrak{c}_1},$	$\frac{b - c \mathfrak{a}_1}{d - a \cdot \mathfrak{b}_1},$	$\frac{a - c \mathfrak{b}_1}{d - b \cdot \mathfrak{a}_1},$

viz. for $x = a$, the relation is $z = y$, but in the other three cases respectively the relation is a linear one, $z = \frac{\alpha y + \beta}{\gamma y + \delta}$.

Rationalising the first equation for $\sqrt{\frac{a-z}{d-z}}$, we have

$$(bc, ad)^2(a - z) = (a - b)(a - c)(d - z) \{ \mathfrak{a} \mathfrak{d} \mathfrak{b}_1 \mathfrak{c}_1 + \mathfrak{a}_1 \mathfrak{d}_1 \mathfrak{b} \mathfrak{c} - 2 \sqrt{\mathfrak{a} \mathfrak{b} \mathfrak{c} \mathfrak{d} \mathfrak{a}_1 \mathfrak{b}_1 \mathfrak{c}_1 \mathfrak{d}_1} \},$$

and thence

$$\{ (bc, ad)^2(a - z) - (a - b)(a - c)(d - z) (\mathfrak{a} \mathfrak{d} \mathfrak{b}_1 \mathfrak{c}_1 + \mathfrak{a}_1 \mathfrak{d}_1 \mathfrak{b} \mathfrak{c}) \}^2 = (a - b)^2(a - c)^2(d - z)^2 \cdot 4 \mathfrak{a} \mathfrak{b} \mathfrak{c} \mathfrak{d} \mathfrak{a}_1 \mathfrak{b}_1 \mathfrak{c}_1 \mathfrak{d}_1.$$

Expanding, and observing that

$$(\mathfrak{a}\mathfrak{d}\mathfrak{b}_1\mathfrak{c}_1 + \mathfrak{a}_1\mathfrak{d}_1\mathfrak{b}\mathfrak{c})^2 - 4\mathfrak{a}\mathfrak{b}\mathfrak{c}\mathfrak{d}\mathfrak{a}_1\mathfrak{b}_1\mathfrak{c}_1\mathfrak{d}_1 = (\mathfrak{a}\mathfrak{d}\mathfrak{b}_1\mathfrak{c}_1 - \mathfrak{a}_1\mathfrak{d}_1\mathfrak{b}\mathfrak{c})^2 = (bc, ad)^2(x-y)^2,$$

the whole equation becomes divisible by $(bc, ad)^2$, and omitting this factor, the equation is

$$(bc, ad)^2(a-z)^2 - 2(a-b)(a-c)(a-z)(d-z)(\mathfrak{a}\mathfrak{d}\mathfrak{b}_1\mathfrak{c}_1 + \mathfrak{a}_1\mathfrak{d}_1\mathfrak{b}\mathfrak{c}) + (a-b)^2(a-c)^2(d-z)^2(x-y)^2 = 0,$$

or, as this may also be written,

$$\begin{aligned} & z^2 \{(bc, ad)^2 - 2(a-b)(a-c)(\mathfrak{a}\mathfrak{d}\mathfrak{b}_1\mathfrak{c}_1 + \mathfrak{a}_1\mathfrak{d}_1\mathfrak{b}\mathfrak{c}) + (a-b)^2(a-c)^2(x-y)^2\} \\ & - 2z \{(bc, ad)^2 a - (a-b)(a-c)(\mathfrak{a}\mathfrak{d}\mathfrak{b}_1\mathfrak{c}_1 + \mathfrak{a}_1\mathfrak{d}_1\mathfrak{b}\mathfrak{c})(a+d) + (a-b)^2(a-c)^2(x-y)^2 d\} \\ & + \{(bc, ad)^2 a^2 - 2(a-b)(a-c)(\mathfrak{a}\mathfrak{d}\mathfrak{b}_1\mathfrak{c}_1 + \mathfrak{a}_1\mathfrak{d}_1\mathfrak{b}\mathfrak{c})ad + (a-b)^2(a-c)^2(x-y)^2 d^2\} = 0. \end{aligned}$$

This is really a symmetrical equation in x, y, z of the form

$$\begin{aligned} & A \\ & + 2B(x+y+z) \\ & + C(x^2+y^2+z^2) \\ & + 2D(yz+zx+xy) \\ & + 2E(y^2z+yz^2+z^2x+zx^2+x^2y+xy^2) \\ & + 4Fxyz \\ & + 2G(x^2yz+xy^2z+xyz^2) \\ & + H(y^2z^2+z^2x^2+x^2y^2) \\ & + 2I(x^2y^2z+x^2yz^2+xy^2z^2) \\ & + Jx^2y^2z^2 = 0; \end{aligned}$$

the several coefficients being symmetrical as regards b, c, d , but the a entering unsymmetrically: the actual values are

$$\begin{aligned} A &= a^4 \{b^2c^2 + b^2d^2 + c^2d^2 - 2bcd(b+c+d)\} + 2a^3bcd(bc+bd+cd) - 3a^2b^2c^2d^2, \\ B &= 2a^4bcd - a^3(b^2c^2 + b^2d^2 + c^2d^2) + ab^2c^2d^2, \\ C &= -4a^3bcd + a^2(bc+bd+cd)^2 - 2abcd(bc+bd+cd) + b^2c^2d^2, \\ D &= -a^4(bc+bd+cd) + a^3(b^2c + bc^2 + b^2d + bd^2 + c^2d + cd^2 - 2bcd) \\ &\quad + a^2 \{b^2c^2 + b^2d^2 + c^2d^2 - bcd(b+c+d)\} - b^2c^2d^2, \\ E &= a^3(bc+bd+cd) - a^2(b^2c + bc^2 + b^2d + bd^2 + c^2d + cd^2) + abcd(b+c+d), \\ F &= a^4(b+c+d) - a^3(b^2 + c^2 + d^2 + bc+bd+cd) + 6a^2bcd \\ &\quad + a \{b^2c^2 + b^2d^2 + c^2d^2 - bcd(b+c+d)\} - bcd(bc+bd+cd), \\ G &= -a^4 + a^2(b^2 + c^2 + d^2 - bc - bd - cd) + a(b^2c + bc^2 + b^2d + bd^2 + c^2d + cd^2 - 2bcd) - bcd(b+c+d), \\ H &= a^4 - 2a^3(b+c+d) + a^2(b+c+d)^2 - 4abcd, \\ I &= a^2 - a(b^2 + c^2 + d^2) + 2bcd, \\ J &= -3a^2 + 2a(b+c+d) + b^2 + c^2 + d^2 - 2(bc+bd+cd). \end{aligned}$$

It may be remarked by way of verification that the equation remains unaltered on substituting for x, y, z, a, b, c, d their reciprocals and multiplying the whole by $a^4 b^4 c^4 d^4 x^2 y^2 z^2$.

I further remark that, writing $a = 0$, we have

$$\begin{aligned} A = 0, \quad B = 0, \quad C = b^2 c^2 d^2, \quad D = -b^2 c^2 d^2, \quad E = 0, \quad F = bcd(bc + bd + cd), \\ G = -bcd(b + c + d), \quad H = 0, \quad I = 2bcd, \quad J = b^2 + c^2 + d^2 - 2(bc + bd + cd); \end{aligned}$$

and writing also $\alpha = 1, -\beta = (b + c + d), \gamma = bc + bd + cd, -\delta = bcd$, (whence

$$a - x \cdot b - x \cdot c - x \cdot d - x = \beta x + \gamma x^2 + \delta x^3 + \varepsilon x^4,)$$

we have the formula

$$\begin{aligned} & \beta(x^2 + y^2 + z^2 - 2yz - 2zx - 2xy) \\ & - 4\beta\gamma xyz \\ & - 2\beta\delta xyz(x + y + z) \\ & - 4\beta\varepsilon xyz(yz + zx + xy) \\ & + (\delta^2 - 4\gamma\varepsilon)x^2 y^2 z^2 = 0, \end{aligned}$$

given p. 348 of my *Elliptic Functions* as a particular integral of the differential equation when the radical is $\sqrt{\beta x + \gamma x^2 + \delta x^3 + \varepsilon x^4}$.

Let the equation in (x, y, z) be called $u = 0$; u has been given in the form $u = \mathfrak{P}z^2 - 2\mathfrak{Q}z + \mathfrak{R}$, and we thence have $\frac{du}{dz} = 0 = -\mathfrak{Q} + \mathfrak{P}z$; which, in virtue of the equation $u = 0$ itself, becomes $\mathfrak{Q}^2 - \mathfrak{P}\mathfrak{R} = 0$. We have easily

$$\mathfrak{Q}^2 - \mathfrak{P}\mathfrak{R} = (a - b)^2(a - c)^2(a - d)^2 \{(\mathfrak{a}\mathfrak{d}\mathfrak{b}_1\mathfrak{c}_1 - \mathfrak{a}_1\mathfrak{d}_1\mathfrak{b}\mathfrak{c})^2 - (bc, ad)^2(x - y)^2\},$$

or, attending to the relation

$$(\mathfrak{a}\mathfrak{d}\mathfrak{b}_1\mathfrak{c}_1 - \mathfrak{a}_1\mathfrak{d}_1\mathfrak{b}\mathfrak{c})^2 = (\mathfrak{a}\mathfrak{d}\mathfrak{b}_1\mathfrak{c}_1 + \mathfrak{a}_1\mathfrak{d}_1\mathfrak{b}\mathfrak{c})^2 - 4\mathfrak{a}\mathfrak{b}\mathfrak{c}\mathfrak{d}\mathfrak{a}_1\mathfrak{b}_1\mathfrak{c}_1\mathfrak{d}_1,$$

this is

$$= (bc, ad)^2(x - y)^2 - 4\mathfrak{a}\mathfrak{b}\mathfrak{c}\mathfrak{d}\mathfrak{a}_1\mathfrak{b}_1\mathfrak{c}_1\mathfrak{d}_1,$$

or we have

$$\mathfrak{Q}^2 - \mathfrak{P}\mathfrak{R} = -4(a - b)^2(a - c)^2(a - d)^2\mathfrak{a}\mathfrak{b}\mathfrak{c}\mathfrak{d}\mathfrak{a}_1\mathfrak{b}_1\mathfrak{c}_1\mathfrak{d}_1.$$

Writing

$$\begin{aligned} \frac{1}{2} \frac{du}{dx} &= (a - b)(a - c)(a - d)\sqrt{abcd} \sqrt{\mathfrak{a}_1\mathfrak{b}_1\mathfrak{c}_1\mathfrak{d}_1}, \\ a - z, \quad b - z, \quad c - z, \quad d - z &= \mathfrak{a}_2, \quad \mathfrak{b}_2, \quad \mathfrak{c}_2, \quad \mathfrak{d}_2, \end{aligned}$$

we have of course the like formulae

$$\begin{aligned} \frac{1}{2} \frac{du}{dy} &= (a - b)(a - c)(a - d)\sqrt{abcd} \sqrt{\mathfrak{a}_2\mathfrak{b}_2\mathfrak{c}_2\mathfrak{d}_2}, \\ \frac{1}{2} \frac{du}{dz} &= (a - b)(a - c)(a - d)\sqrt{abcd} \sqrt{\mathfrak{a}\mathfrak{b}\mathfrak{c}\mathfrak{d}\mathfrak{a}_2\mathfrak{b}_2\mathfrak{c}_2\mathfrak{d}_2} \end{aligned}$$

and the equation $du = 0$ then gives

$$\frac{dx}{\sqrt{abcd}\sqrt{abc\mathfrak{d}}} + \frac{dy}{\sqrt{a_1b_1c_1\mathfrak{d}_1}} + \frac{dz}{\sqrt{a_2b_2c_2\mathfrak{d}_2}} = 0,$$

as it should do. The differential equation might also have been verified directly from any one of the expressions for

$$\sqrt{\frac{a-z}{d-z}}, \quad \sqrt{\frac{b-z}{d-z}}, \quad \sqrt{\frac{c-z}{d-z}}.$$

Writing for shortness

$$X = a - x \cdot b - x \cdot c - x \cdot d - x, \text{ etc.,}$$

then the general integral of the differential equation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}} = 0$$

by Abel's theorem is

$$\begin{vmatrix} x^2, & x, & 1, & \sqrt{X} \\ y^2, & y, & 1, & \sqrt{Y} \\ z^2, & z, & 1, & \sqrt{Z} \\ w^2, & w, & 1, & \sqrt{W} \end{vmatrix} = 0,$$

where w is the constant of integration: and it is to be shown that the value of w which corresponds to the integral given in the present paper is $= a$. Observe that writing in the determinant $w = a$, the determinant on putting therein $x = a$, would vanish whether z were or were not $= y$; but this is on account of an extraneous factor $a - w$, so that we do not thus prove the required theorem that (w being $= a$) we have $y = z$ when $x = a$.

An equivalent form of Abel's integral is that there exist values A, B, C such that

$$Ax^2 + Bx + C = \sqrt{X},$$

$$Ay^2 + By + C = \sqrt{Y},$$

$$Az^2 + Bz + C = \sqrt{Z},$$

$$Aw^2 + Bw + C = \sqrt{W},$$

or, what is the same thing, that we have identically

$$-(A\theta^2 + B\theta + C)^2 + \Theta = (A^2 - 1) \cdot \theta - x \cdot \theta - y \cdot \theta - z \cdot \theta - w.$$

We have therefore

$$C^2 - abcd = (A^2 - 1)xyzw,$$

or say

$$xyzw = \frac{C^2 - abcd}{A^2 - 1}.$$

which equation, regarding therein A, B, C as determined by the three equations

$$Ax^2 + Bx + C = \sqrt{X},$$

$$Ay^2 + By + C = \sqrt{Y},$$

$$Aw^2 + Bw + C = \sqrt{W},$$

is a form of Abel's integral, giving z rationally in terms of x, y, w .

Supposing that, when $x = a, z = y$; then the last-mentioned integral gives

$$ay^2w = \frac{C^2 - abcd}{A^2 - 1},$$

where A, C are now determined by the equations

$$Aa^2 + Ba + C = 0,$$

$$Ay^2 + By + C = \sqrt{Y},$$

$$Aw^2 + Bw + C = \sqrt{W},$$

and, imagining these values actually substituted, it is to be shown that the equation

$$ay^2w = \frac{C^2 - abcd}{A^2 - 1}$$

is satisfied by the value $w = a$.

We have

$$A(a - y)(a - w)(w - y) = (a - w)\sqrt{Y} - (a - y)\sqrt{W},$$

$$B(a - y)(a - w)(w - y) = (a - w)(a + w)\sqrt{Y} - (a - y)(a + y)\sqrt{W},$$

$$C(a - y)(a - w)(w - y) = (a - w)aw\sqrt{Y} - (a - y)ay\sqrt{W},$$

or writing as before

$$a - y, b - y, c - y, d - y = \mathfrak{a}_1, \mathfrak{b}_1, \mathfrak{c}_1, \mathfrak{d}_1,$$

and also

$$a - w, b - w, c - w, d - w = \mathfrak{a}_3, \mathfrak{b}_3, \mathfrak{c}_3, \mathfrak{d}_3,$$

then $Y = \mathfrak{a}_1\mathfrak{b}_1\mathfrak{c}_1\mathfrak{d}_1$, $W = \mathfrak{a}_3\mathfrak{b}_3\mathfrak{c}_3\mathfrak{d}_3$, and the formulae become

$$A = \frac{1}{(w - y)\sqrt{\mathfrak{a}_1\mathfrak{a}_3}} \{ \sqrt{\mathfrak{a}_3\mathfrak{b}_1\mathfrak{c}_1\mathfrak{d}_1} - \sqrt{\mathfrak{a}_1\mathfrak{b}_3\mathfrak{c}_3\mathfrak{d}_3} \},$$

$$B = \frac{1}{(w - y)\sqrt{\mathfrak{a}_1\mathfrak{a}_3}} \{ -(a + w)\sqrt{\mathfrak{a}_3\mathfrak{b}_1\mathfrak{c}_1\mathfrak{d}_1} + (a + y)\sqrt{\mathfrak{a}_1\mathfrak{b}_3\mathfrak{c}_3\mathfrak{d}_3} \},$$

$$C = \frac{1}{(w - y)\sqrt{\mathfrak{a}_1\mathfrak{a}_3}} \{ aw\sqrt{\mathfrak{a}_3\mathfrak{b}_1\mathfrak{c}_1\mathfrak{d}_1} - ay\sqrt{\mathfrak{a}_1\mathfrak{b}_3\mathfrak{c}_3\mathfrak{d}_3} \}.$$

If in these formulae w is indefinitely nearly $= a$, then $\mathfrak{a}_3$ is indefinitely small, so that $\sqrt{\mathfrak{a}_3\mathfrak{b}_1\mathfrak{c}_1\mathfrak{d}_1}$ may be neglected in comparison with $\sqrt{\mathfrak{a}_1\mathfrak{b}_3\mathfrak{c}_3\mathfrak{d}_3}$: also $w - y$ may be put $= \mathfrak{a}_1$; the formulae thus become

$$A = -\frac{\sqrt{\mathfrak{b}_3\mathfrak{c}_3\mathfrak{d}_3}}{\mathfrak{a}_1\sqrt{\mathfrak{a}_3}}, \quad B = (a + y) \frac{\sqrt{\mathfrak{b}_3\mathfrak{c}_3\mathfrak{d}_3}}{\mathfrak{a}_1\sqrt{\mathfrak{a}_3}}, \quad C = -ay \frac{\sqrt{\mathfrak{b}_3\mathfrak{c}_3\mathfrak{d}_3}}{\mathfrak{a}_1\sqrt{\mathfrak{a}_3}},$$

where the values of A , B , C are each of them indefinitely large on account of the factor $\sqrt{\mathfrak{a}_3}$ in the denominator; the value of C is $C = ayA$, and substituting this value in the equation

$$ay^2w = \frac{C^2 - abcd}{A^2 - 1},$$

and then considering A as indefinitely large, the equation becomes $ay^2w = a^2y^2$, that is, $w = a$; so that $w = a$ is a value of w satisfying this equation.

Cambridge, 3 July, 1878.

698.

ON A THEOREM RELATING TO COVARIANTS.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXVII. (1878), pp. 82, 83.]

The theorem given by Prof. Sylvester, *Crelle*, vol. LXXXV., p. 109, may be stated as follows: If for a binary quantic of the order i in the variables, we consider the whole system of covariants of the degree j in the coefficients, then

$$\Sigma \theta (k + 1) = \frac{\Pi(i + j)}{\Pi(i) \Pi(j)},$$

where θ denotes the number of aszygetic covariants of the order θ in the variables, the values of θ being $ij, ij - 2, ij - 4, \dots, 1$ or 0 , according as ij is odd or even.

In the case of the binary quintic $= (a, \dots | x, y)^5$, ($i = 5$), we have a series of verifications in the Table 88 of my “Ninth Memoir on Quantics,” *Phil. Trans.*, vol. CLXI. (1871), [462]: viz. writing the small letters $a, b, c, \dots, u, v, w$ (instead of the capitals A, B , etc.) to denote the covariants of the quintic, a , the quintic itself, degree 1, order 5, or as I express it, deg-order 1.5: b , the covariant deg-order 2.2, etc., the whole series of deg-orders being

$a,$	$b,$	$c,$	$d,$	$e,$	$f,$	$g,$	$h,$	$i,$	$j,$	$k,$	$l,$
1.5,	2.2,	2.6,	3.3,	3.5,	3.9,	4.0,	4.4,	4.6,	5.1,	5.3,	5.7,
$m,$	$n,$	$o,$	$p,$	$q,$	$r,$	$s,$	$t,$	$u,$	$v,$	$w,$	
6.2,	6.4,	7.1,	7.5,	8.0,	8.2,	9.3,	11.1,	12.0,	13.1,	18.0,	

then the table shows for each deg-order, the several covariants of that deg-order, and